

**Mutual Fund Performance, Risk & Investor Analytics
Platform**

Final Project Report

Data Analyst Internship Capstone Project

Prepared By: Ashutosh Suryawanshi

Date: June 2026

Table of Contents

1. Executive Summary
 2. Introduction
 3. Problem Statement
 4. Project Objectives
 5. Data Sources
 6. ETL Design and Data Architecture
 7. Data Cleaning and Preprocessing
 8. Exploratory Data Analysis
 9. Performance Analytics
 10. Risk Analytics
 11. Investor Analytics
 12. Fund Scorecard Framework
 13. Recommendation Engine
 14. Power BI Dashboard
 15. Key Findings
 16. Limitations
 17. Recommendations
 18. Conclusion
 19. References
-

1. Executive Summary

This project presents the development of a Mutual Fund Performance, Risk and Investor Analytics Platform designed to analyze mutual fund datasets and generate actionable business insights. The platform integrates multiple datasets including NAV history, fund master data, benchmark indices, portfolio holdings, and investor transactions.

The project performs end-to-end data processing, exploratory data analysis, performance evaluation, risk assessment, investor behavior analysis, and dashboard reporting. Key analytical techniques include Alpha-Beta analysis, Sharpe Ratio calculation, Value at Risk (VaR), Conditional Value at Risk (CVaR), Cohort Analysis, SIP Continuity Analysis, and fund recommendation scoring.

The final solution provides investment insights through an interactive Power BI dashboard and supports data-driven decision making for fund managers and investors.

2. Introduction

Mutual funds generate large amounts of financial and transactional data. Effective analysis of this data is essential for evaluating fund performance, understanding investor behavior, and managing portfolio risk.

The objective of this project is to build a comprehensive analytics platform capable of transforming raw mutual fund datasets into meaningful business insights.

3. Problem Statement

Financial institutions face challenges in:

- Tracking fund performance consistently
- Measuring risk-adjusted returns
- Understanding investor retention
- Comparing schemes against benchmarks
- Presenting insights in an accessible format

This project addresses these challenges through data analytics and visualization.

4. Project Objectives

The key objectives were:

- Collect and integrate mutual fund datasets
 - Perform data cleaning and preprocessing
 - Build a centralized SQLite database
 - Analyze fund performance
 - Evaluate risk metrics
 - Study investor behavior
 - Develop a recommendation engine
 - Create a Power BI dashboard
-

5. Data Sources

The project utilizes ten datasets:

1. Fund Master Data
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

These datasets provide comprehensive coverage of fund operations, performance, and investor activity.

6. ETL Design and Data Architecture

The ETL pipeline follows the workflow:

Raw CSV Files

- Data Cleaning
- Data Validation
- SQLite Database
- Python Analytics
- Power BI Dashboard

The database acts as the central repository for analytics and reporting.

7. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

- Missing value handling
- Duplicate removal
- Date standardization
- Data type correction
- Feature engineering
- Consistency validation

Output datasets include:

- clean_nav.csv
 - clean_transactions.csv
 - clean_performance.csv
-

8. Exploratory Data Analysis

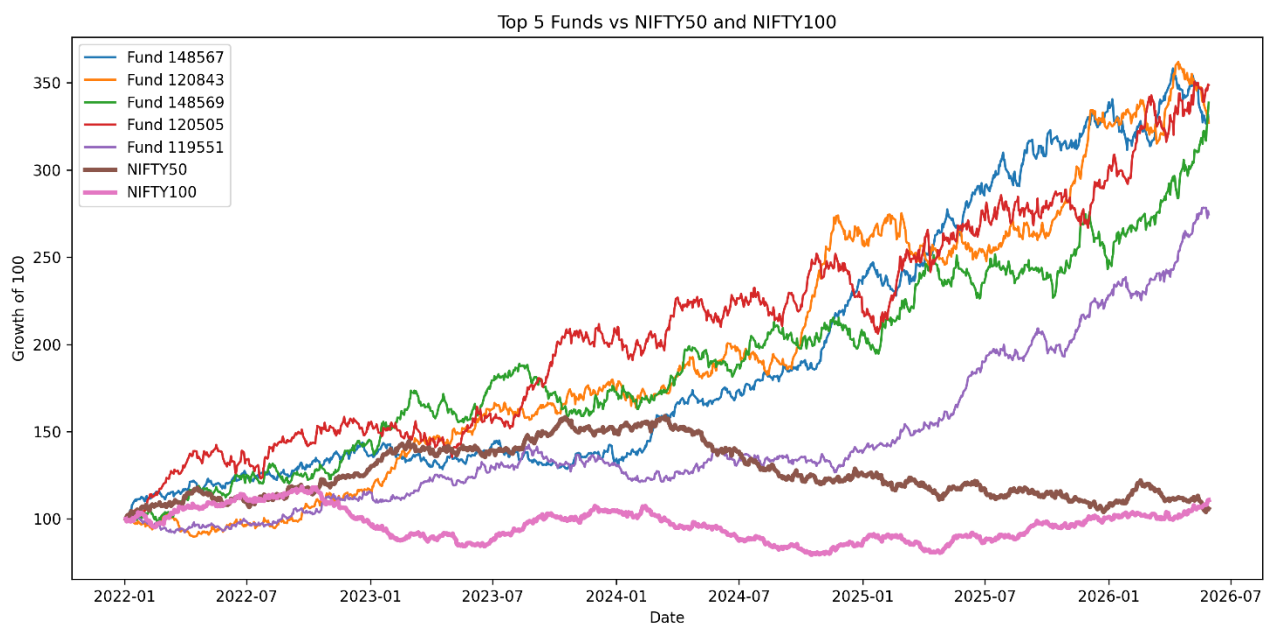
EDA was conducted to understand:

- Fund distribution
- Investment patterns
- Transaction behavior
- Benchmark movement

- Portfolio characteristics

Key observations revealed variations in fund performance across categories and differences in investor activity levels.

9. Performance Analytics



Performance evaluation was conducted using:

Alpha

Measures excess return relative to the benchmark.

Beta

Measures sensitivity to market movements.

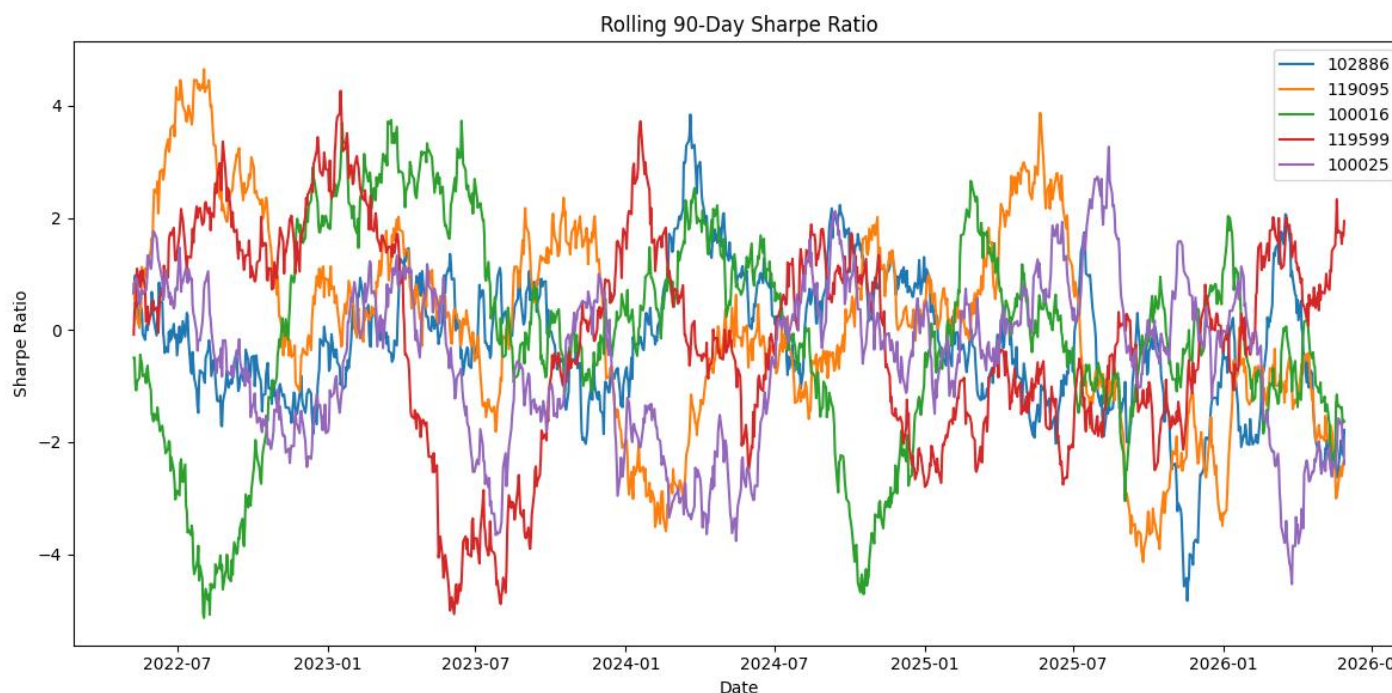
Sharpe Ratio

Measures risk-adjusted returns.

Benchmark Comparison

Compares mutual fund returns against market indices.
Results were stored in alpha_beta.csv and visualized through benchmark comparison charts.

10. Risk Analytics



Risk evaluation included:

Value at Risk (VaR)

Estimates potential portfolio loss under normal market conditions.

Conditional VaR (CVaR)

Measures expected loss during extreme market events.

HHI Concentration Analysis

Evaluates portfolio diversification and concentration risk.

Outputs:

- var_cvar_report.csv
 - hhi_concentration_report.csv
-

11. Investor Analytics

Cohort Analysis

Groups investors based on first investment period and tracks retention patterns.

SIP Continuity Analysis

Measures consistency of recurring SIP investments.

These analyses provide insights into investor engagement and long-term retention.

12. Fund Scorecard Framework

A fund scorecard was developed to rank funds based on:

- Performance
- Risk-adjusted returns
- Benchmark comparison
- Stability metrics

Output:

fund_scorecard.csv

13. Recommendation Engine

A Python-based recommendation engine was developed.

Features:

- Fund ranking
- Risk-adjusted scoring
- Investment recommendations

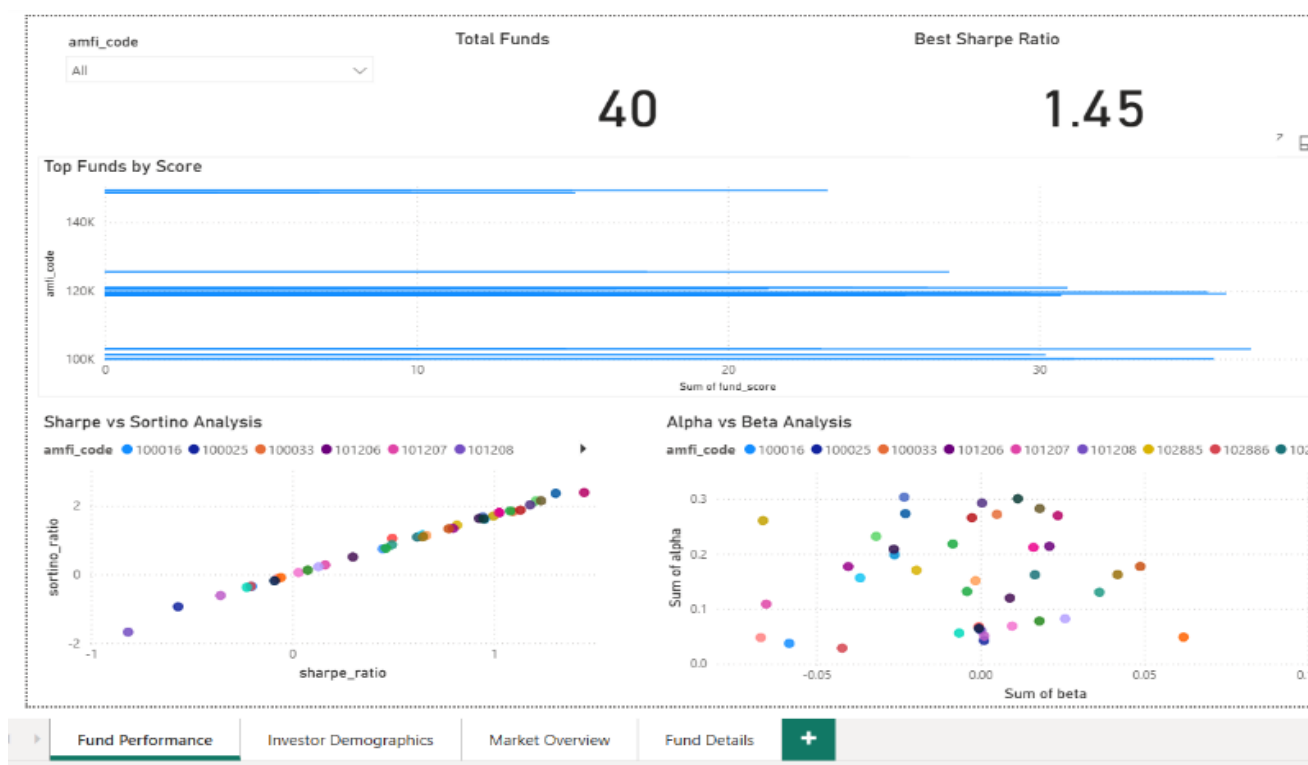
Implementation:

scripts/recommender.py

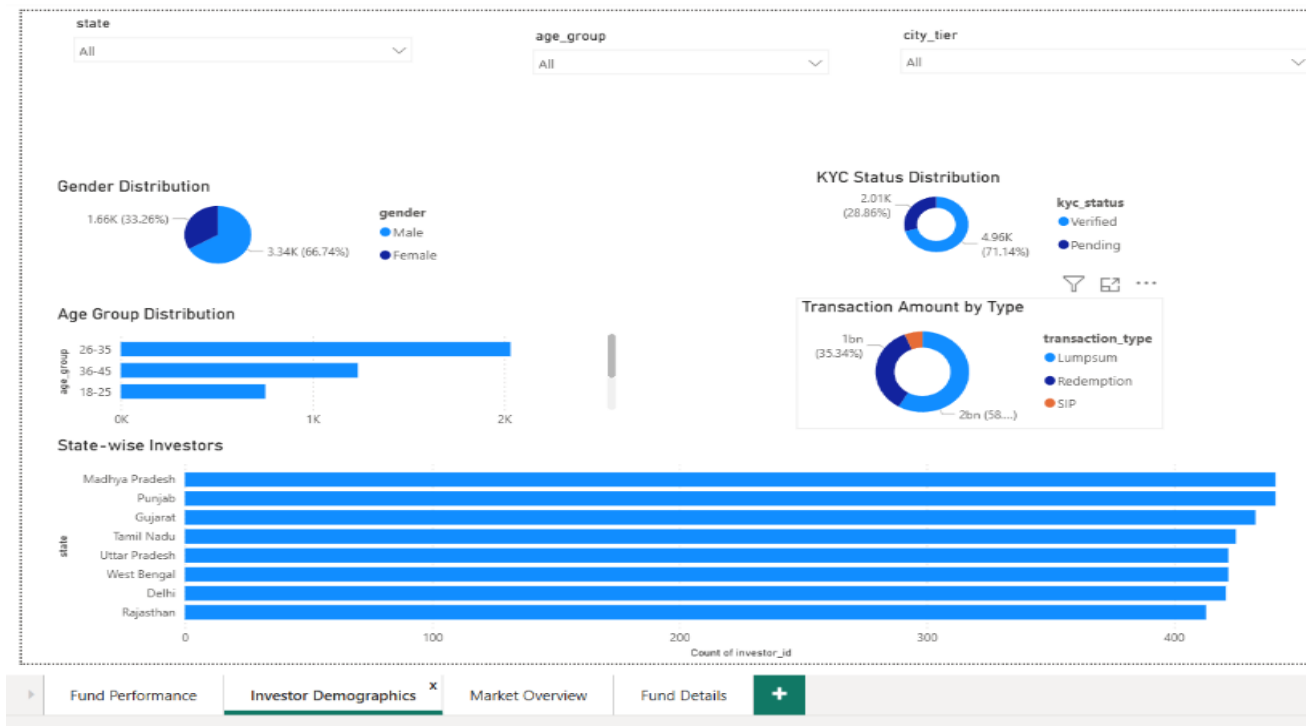
14. Power BI Dashboard

The project includes an interactive Power BI dashboard with four pages:

- Fund Performance



- Investor Demographics



• Market Overview



• Fund Details

amfi_code	alpha	beta
148567	0.27	0.02

amfi_code

All

amfi_code	cagr	sharpe_ratio	sortino_ratio	max_drawdown	fund_score
148567	0.31	1.45	2.39	-0.11	3.60

Fund Details

Fund Performance

Investor Demographics

Market Overview

Fund Details

15. Key Findings

- Strong variation exists in fund performance across categories.
- Risk-adjusted metrics provide better evaluation than raw returns.
- Investor retention declines over time.
- SIP continuity is associated with long-term engagement.
- Dashboard reporting improves business visibility.

16. Limitations

- Historical data only
 - No live market integration
 - Limited predictive modeling
 - Simplified recommendation logic
-

17. Recommendations

- Integrate real-time NAV feeds
 - Implement machine learning models
 - Enhance investor segmentation
 - Deploy cloud-based reporting
 - Add predictive risk analytics
-

18. Conclusion

The project successfully developed a comprehensive Mutual Fund Performance, Risk and Investor Analytics Platform using Python, SQLite and Power BI. The platform integrated multiple datasets, performed advanced analytics, measured investment risk, evaluated investor retention patterns and delivered actionable insights through interactive dashboards. The project demonstrates practical applications of data engineering, financial analytics and business intelligence in the mutual fund industry and provides a strong foundation for future predictive analytics and real-time reporting solutions.

19. References

- [AMFI Mutual Fund Data](#)
- [Benchmark Index Data](#)
- [Power BI Documentation](#)
- [Python Pandas Documentation](#)
- [SQLite Documentation](#)